



Chapter 14

DESCRIPTION OF MOTION (PART TWO)

Evaluation

Convert:

5 h = min

2 h = s

7200 s = min

15 min = h

Evaluation

1) Establish the relationship: $1 \text{ m/s} = 3.6 \text{ km/h}$

2) Convert

- $20 \text{ m/s} = \quad \quad \quad \text{km/h}$

- $216 \text{ km/h} = \quad \quad \quad \text{m /s}$

List of Terms

(1) Average Speed

(2) Instantaneous Speed

(3) Velocity Vector

Evaluation

Example 14.2

At a certain instant, the abscissa of a moving particle with respect to the frame of reference $(O, \vec{i})$ is $x = 3$ (Figure 14.6). Determine its abscissa with respect to $(O_1, \vec{i})$ such that $\overline{O_1O} = 2$.

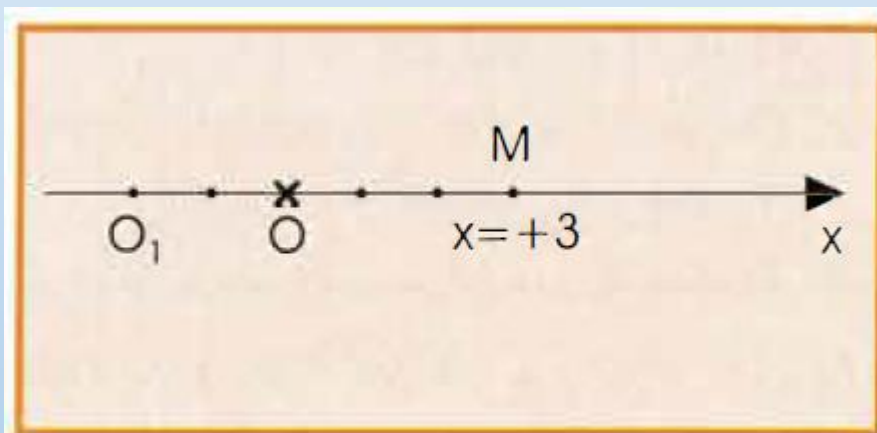


Figure 14.6.

(1) Average Speed

d: distance traveled

Δt : time taken to travel distance d

$$\text{average speed} = \frac{\text{distance traveled}}{\text{time taken}}$$

$$v_{av} = \frac{d}{\Delta t}$$

In the SI system, d is measured in meters (m) and Δt in seconds (s), then v_{av} is measured in meters / second (m/s or m.s^{-1}).

Evaluation

Exemple: A car takes 2 h to travel a distance of 150 km. Calculate the car's average speed.

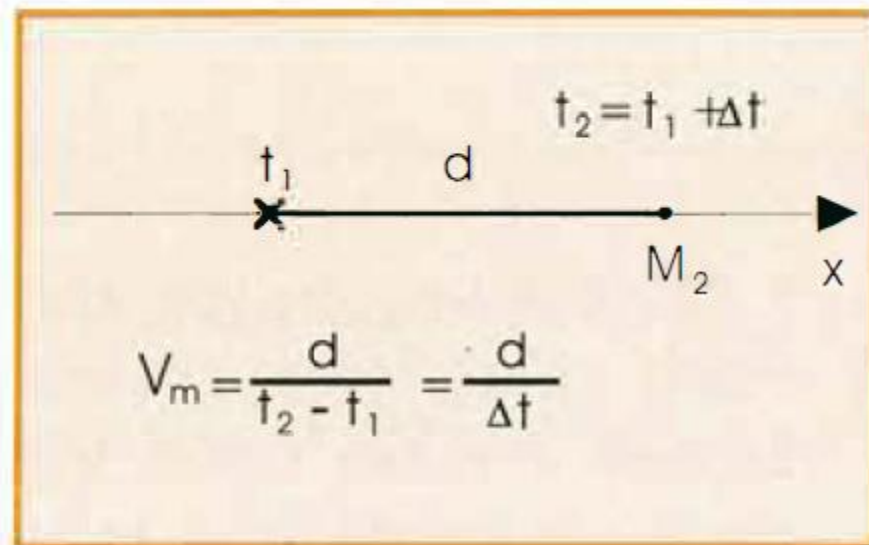


Figure 14.7. $V_a = \frac{d}{\Delta t}$

(2) Instantaneous Speed

The reading of the speedometer at any instant of time represents the instantaneous speed.



(2) Instantaneous Speed

- The instantaneous speed v at an instant t is approximately equal to the average speed between two instants t_1 and t_2 very close to t , with t lying between t_1 and t_2 ($t_1 < t < t_2$).
- The smaller the time interval $\Delta t = t_2 - t_1$, the closer the average speed is to the instantaneous speed at time t .
- The instantaneous speed is also measured, in SI units, in m/s or in $\text{m}\cdot\text{s}^{-1}$.

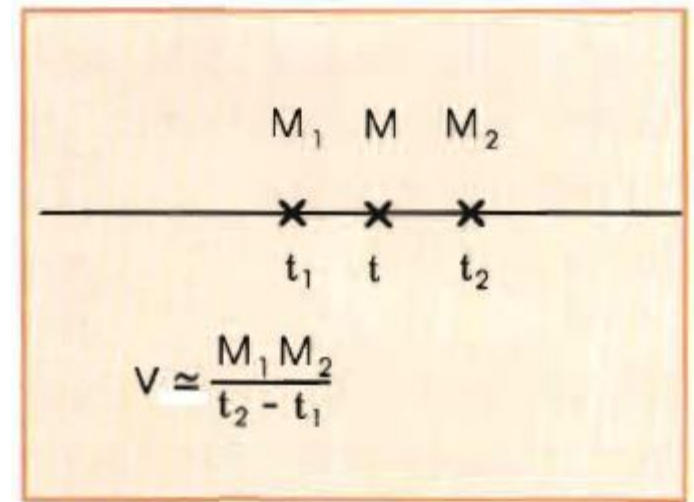


Figure 14.8. Plus l'intervalle de temps $\Delta t = t_2 - t_1$ est petit, plus V est proche de V_m .

Evaluation

Example 14.3

Figure 14.9 shows the motion of a puck, left without initial speed on an inclined air table. The time interval between two consecutive dots is $\tau = 60 \text{ ms}$.

- Determine the average speed of the puck between t_0 and t_3 , t_1 and t_4 , and t_2 and t_6 .
- Determine the instantaneous speeds of the puck at the instants t_1 , t_2 , and t_3 . What can you deduce?

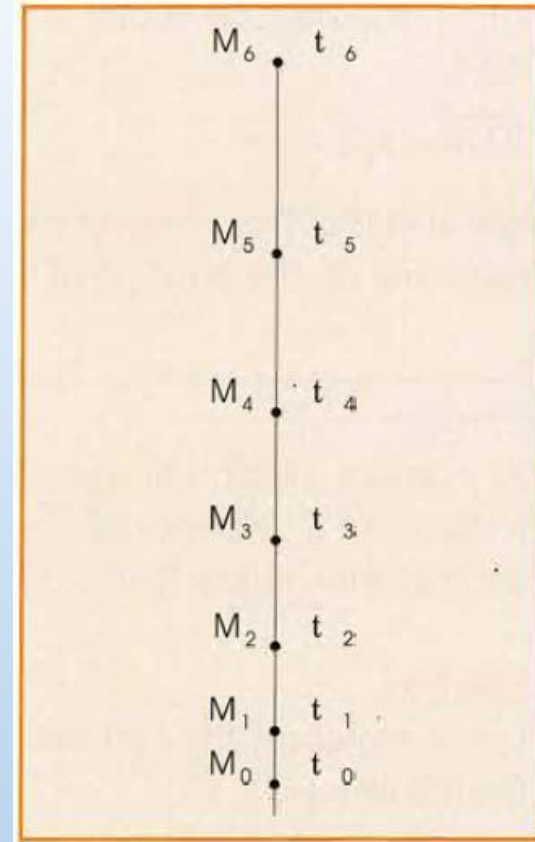


Figure 14.9. Recording of the motion of the puck.

Solution

a. The distances traveled by the puck during the time intervals $t_3 - t_0 = 3 \tau = 3 \times 60 \text{ ms}$, $t_4 - t_2 = 2 \times 60 \text{ ms}$ and $t_6 - t_2 = 4 \times 60 \text{ ms}$ are respectively:

$$M_0M_3 = 23 \text{ mm}, M_2M_4 = 22 \text{ mm} \text{ and } M_2M_6 = 55 \text{ mm}$$

The average speeds during these time intervals are respectively:

$$V_{0,3} = 23 / 180 = 0.128 \text{ mm / ms},$$

$$V_{2,4} = 22 / 120 = 0.183 \text{ mm / ms} \text{ and}$$

$$V_{2,6} = 55 \text{ mm} / 240 \text{ ms} = 0.229 \text{ mm / ms}$$

b. The instantaneous speeds at instants t_1 , t_2 and t_3 are:

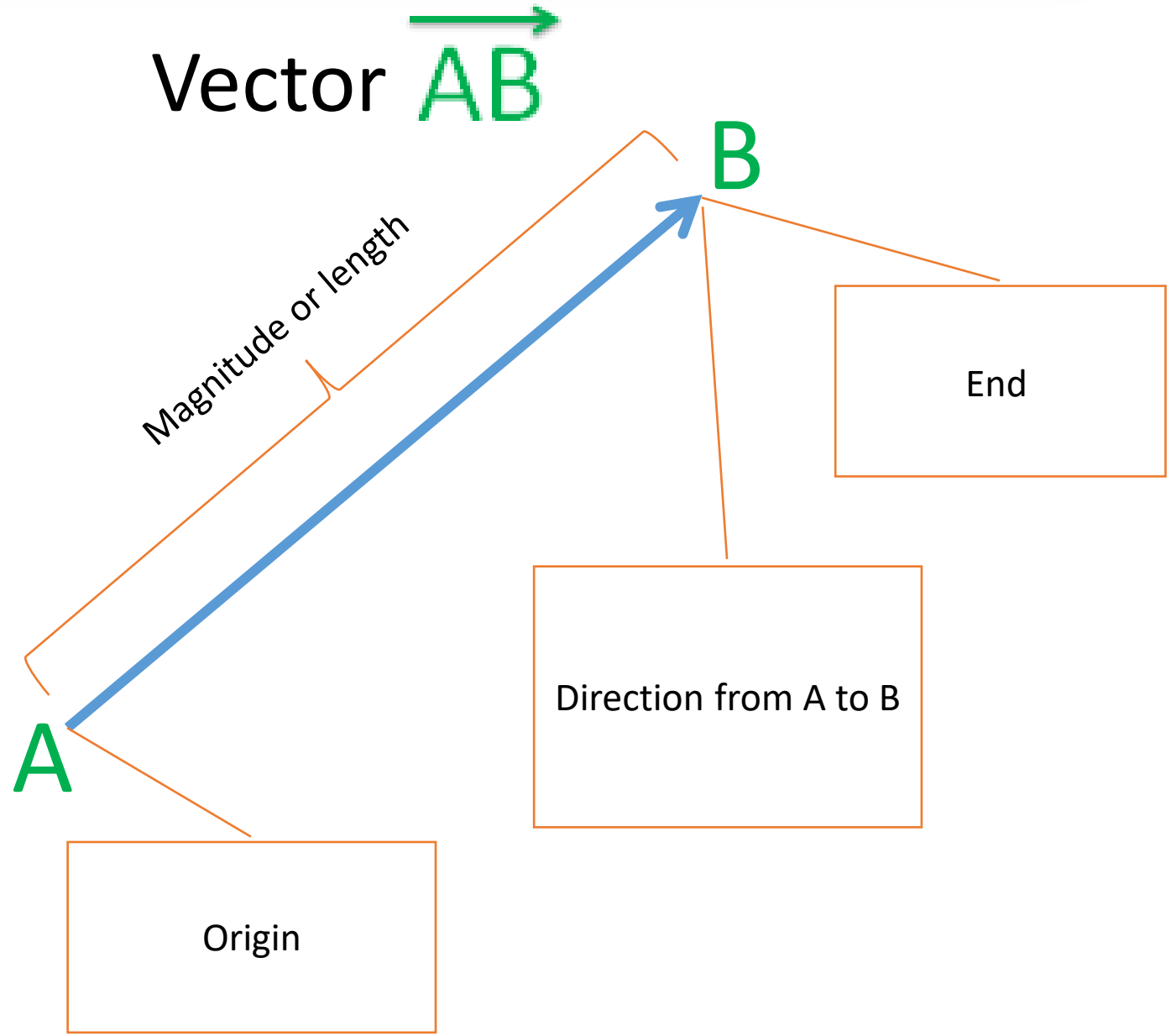
$$V_1 \cong V_{0,2} = M_0M_2 / (t_2 - t_0) = 13 / (2 \times 60) = 0.108 \text{ mm / ms}$$

$$V_2 \cong V_{1,3} = M_1M_3 / (t_3 - t_1) = 18 / (2 \times 60) = 0.150 \text{ mm / ms}$$

$$V_3 \cong V_{2,4} = M_2M_4 / (t_4 - t_2) = 22 / (2 \times 60) = 0.183 \text{ mm / ms}$$

We deduce that v is increasing.

Vectors



Characteristics of Vectors

POINT

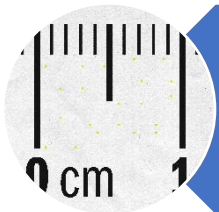
Origin: point O.



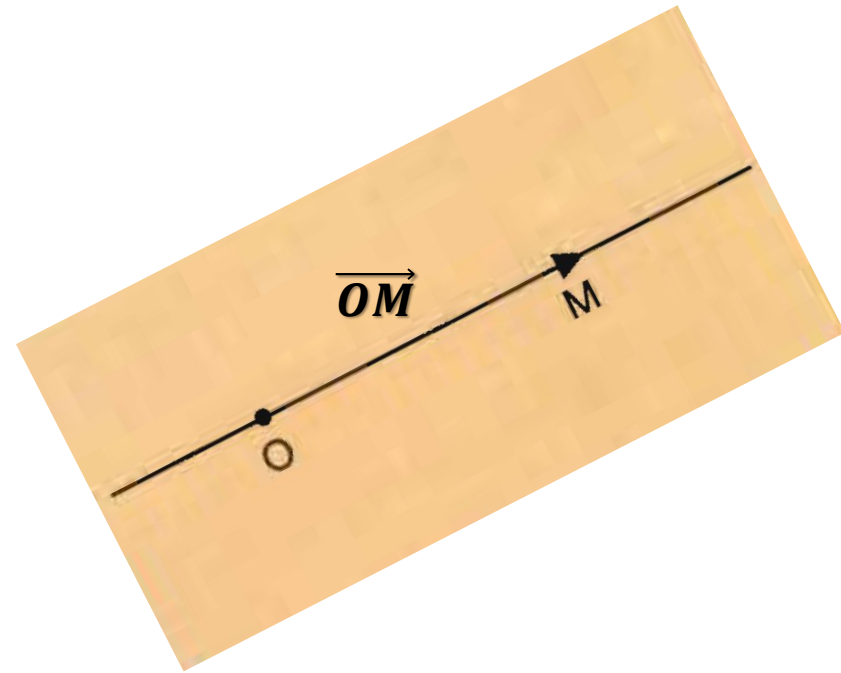
Line of action: straight line (OM).



Direction: from O to M.



Magnitude: length of segment [OM].



(3) Velocity Vector



(3) Velocity Vector

- To describe the motion of a mobile at an instant t , it is not enough to know the value of the **instantaneous speed**, but also the **position of the mobile** and the **direction of motion** must be known.
- The velocity vector $\vec{v}$ has the following elements:
 - ☐ Origin : the position of the object at the considered instant.
 - ☐ Direction: that of the motion.
 - ☐ Line of action : the rectilinear trajectory.
 - ☐ Magnitude: the speed at that instant.

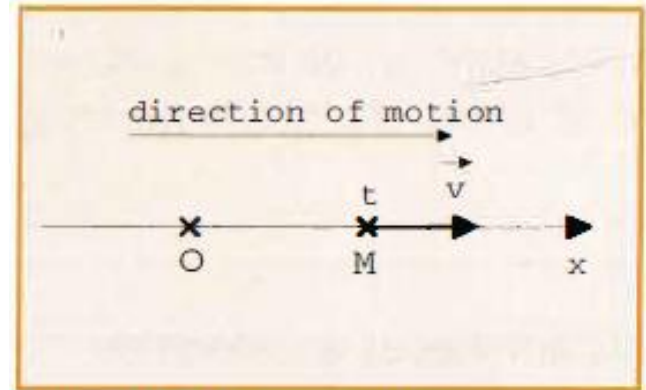


Figure 14.10. Representation of the velocity vector. The chosen scale is 1 cm to 5m/s. A speed of 5m/s is represented by a vector of length 1cm.

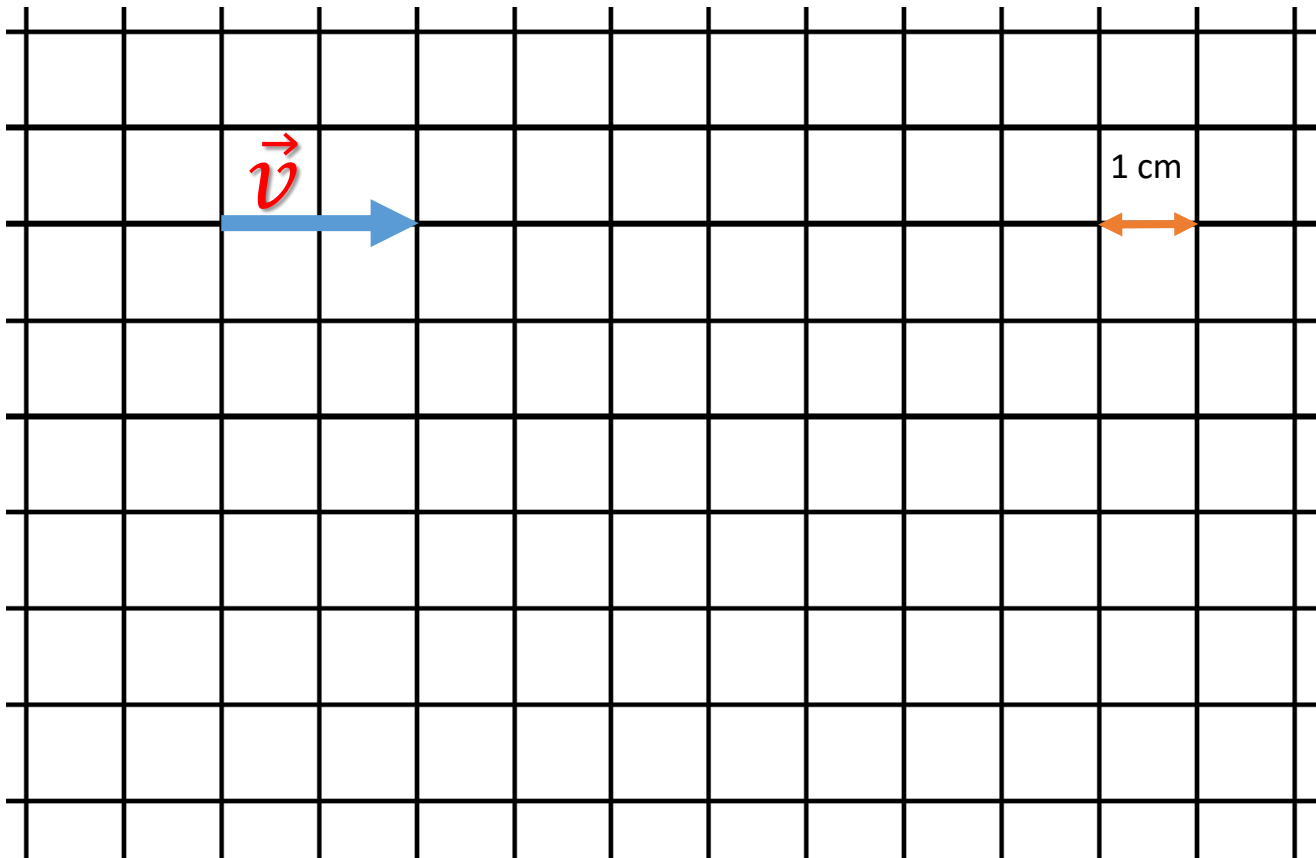
Note

The length of the arrow, which represents the velocity vector $\vec{v}$, depends on:

1. The value of speed
2. The scale chosen

Evaluation

The velocity vector $\vec{v}$, shown below, is represented at the scale 10 m/s per centimeter. Represent the same vector using the scale: 5 m/s / 1 cm.



Evaluation

Example 14.4

A train moves on rectilinear rails. At an instant t , it passes by the midpoint M of its path and its velocity can be represented by the vector $\overrightarrow{MN}$ of length $MN = 2 \text{ cm}$ drawn to the scale $1 \text{ cm to } 10 \text{ m/s}$ (Figure 14.11).

Determine, at instant t , the direction of the motion and the magnitude of the velocity in km/h .

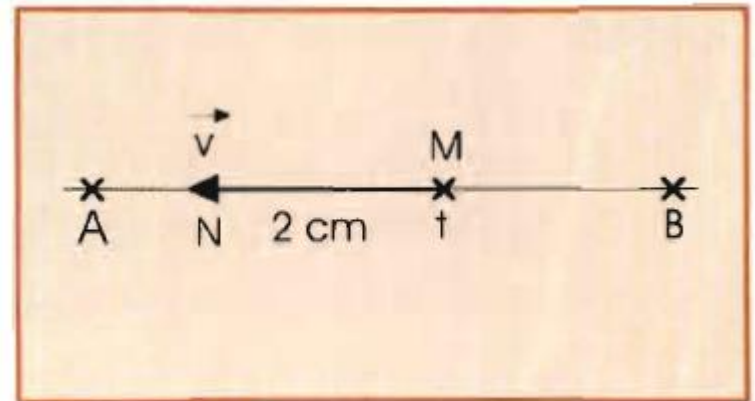


Figure 14.11.

Solution

The direction of the motion at instant t is that of its velocity at the same instant t , (from B to A).

Referring to the chosen scale, the magnitude of the velocity is:

$$v = 2 \text{ cm} \times 10 \text{ m/s} / 1 \text{ cm} = 20 \text{ m/s}$$

Since $1 \text{ m/s} = 10^{-3} \text{ km} \times 3600/\text{h} = 3.6 \text{ km/h}$

$$v = 20 \times 3.6 = 72 \text{ km/h}$$